

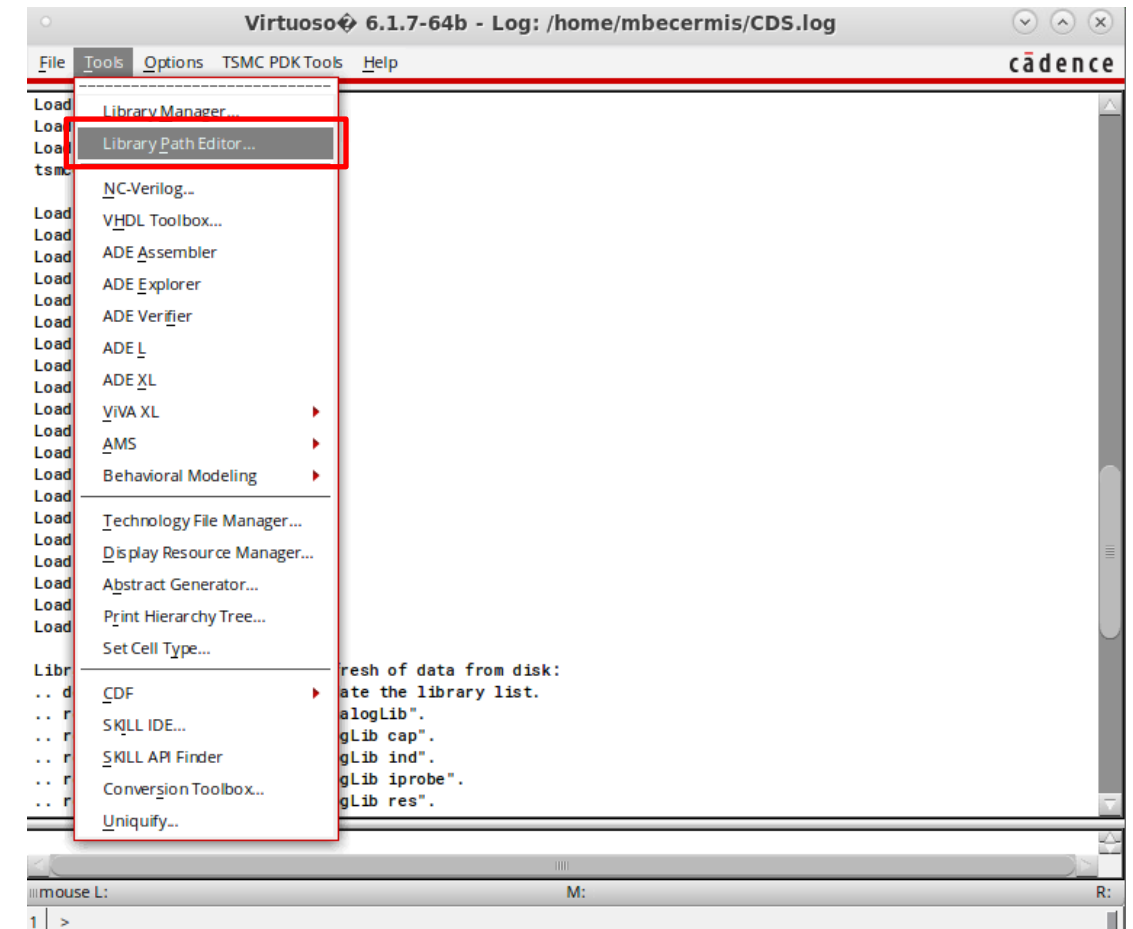
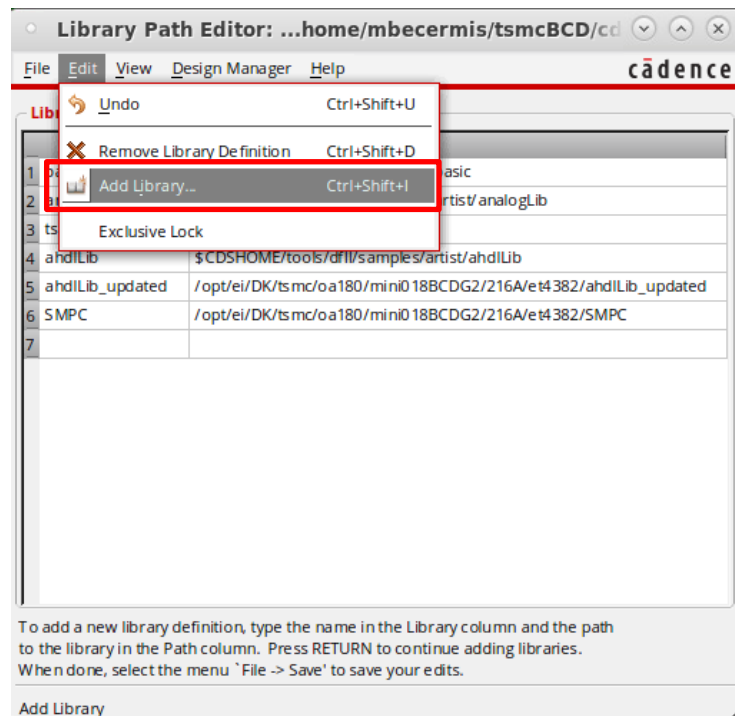
ET4382: Power Conversion Techniques

Tutorial 4

4th March, 2021

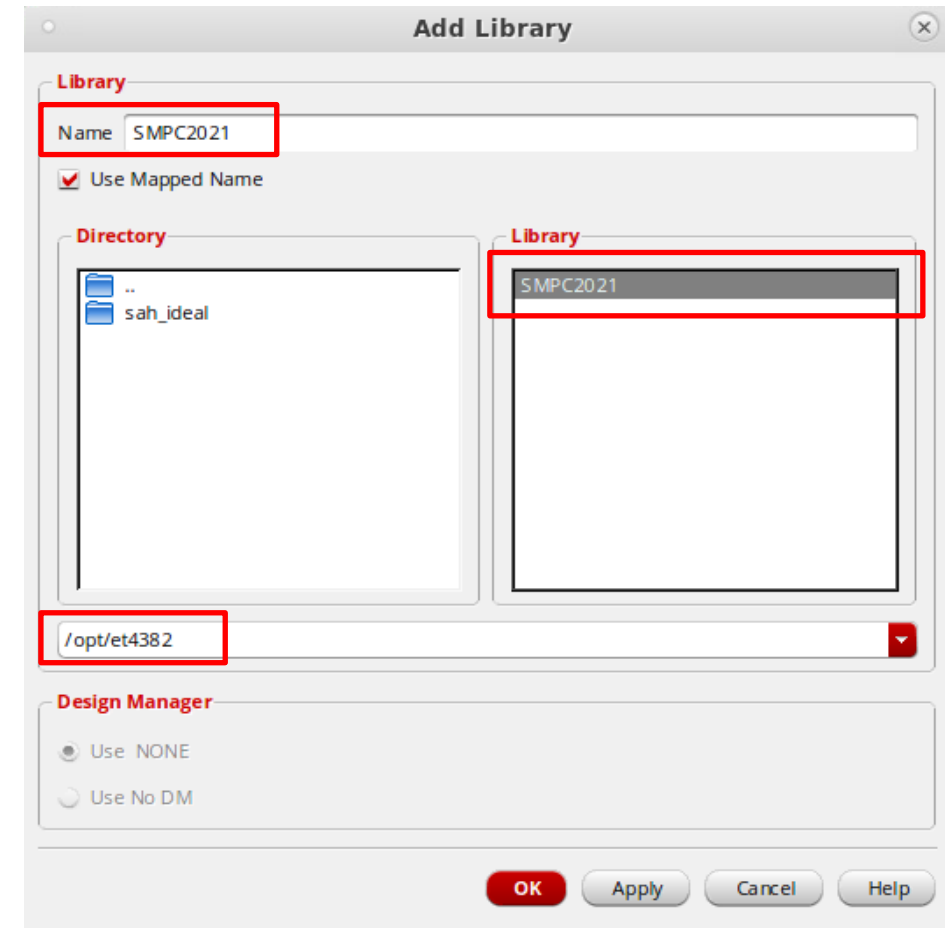
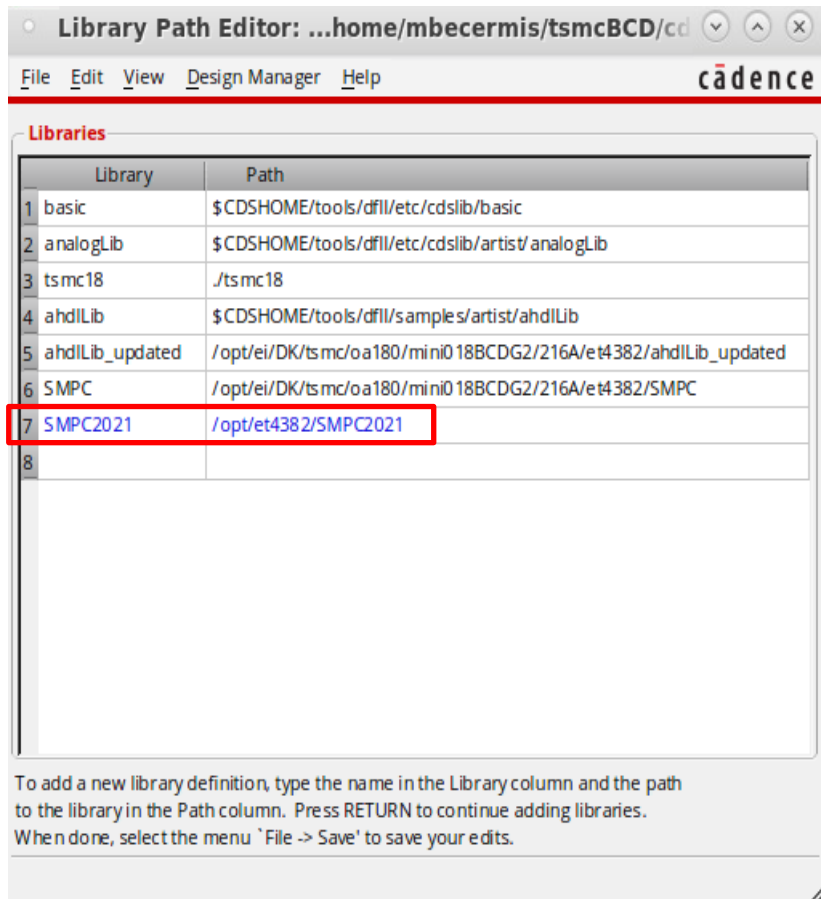
Adding a new library

- **This is required for homework 3-4!** You will find schematics and testbenches of homework 3-4 in SMPC2021 library.
- You can add “SMPC2021” to Library Manager through Library Path Editor.
- Go to **Tools -> Library Path Editor** in the CIW
- Go to **Edit -> Add Library** in the Library Path Editor



Adding a new library

- Find the following directory:
/opt/et4382
- Select SMPC2021 library
- Check the library name is SMPC 2021, and click OK

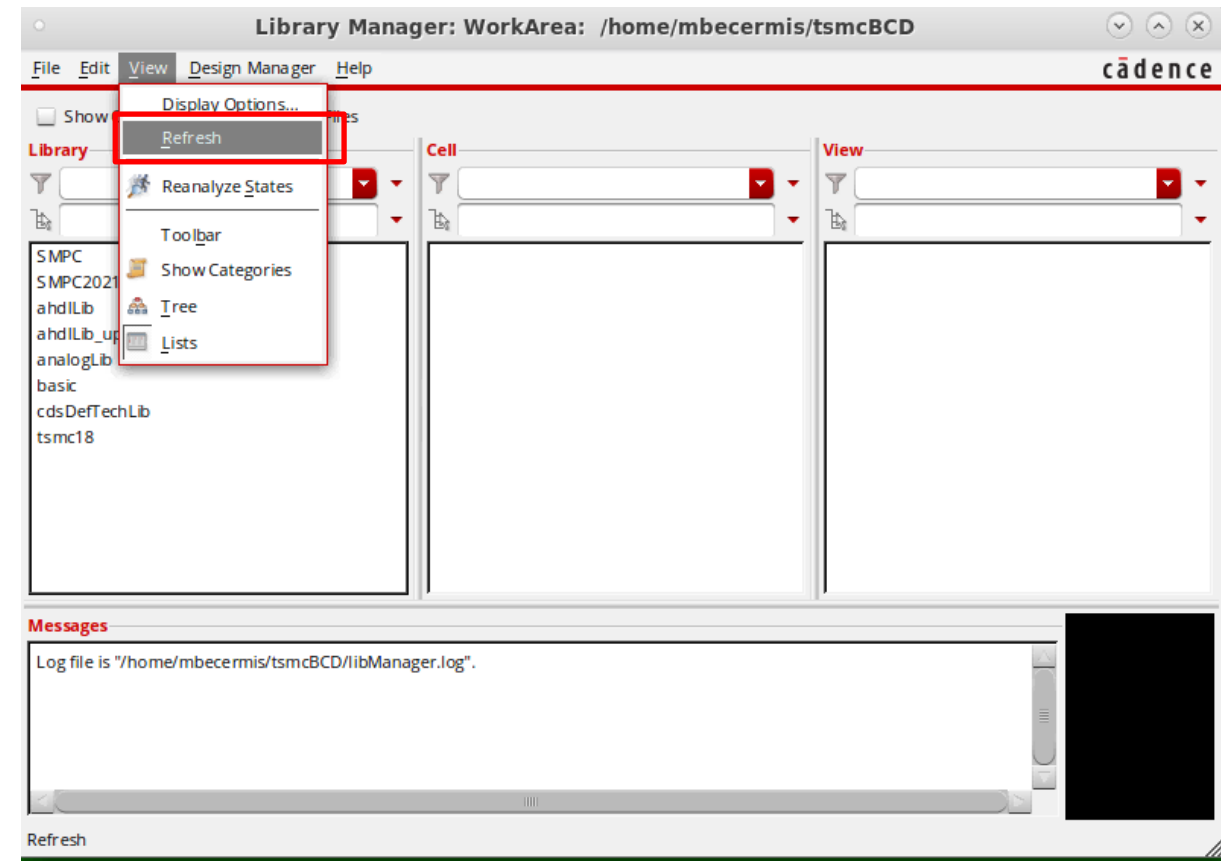
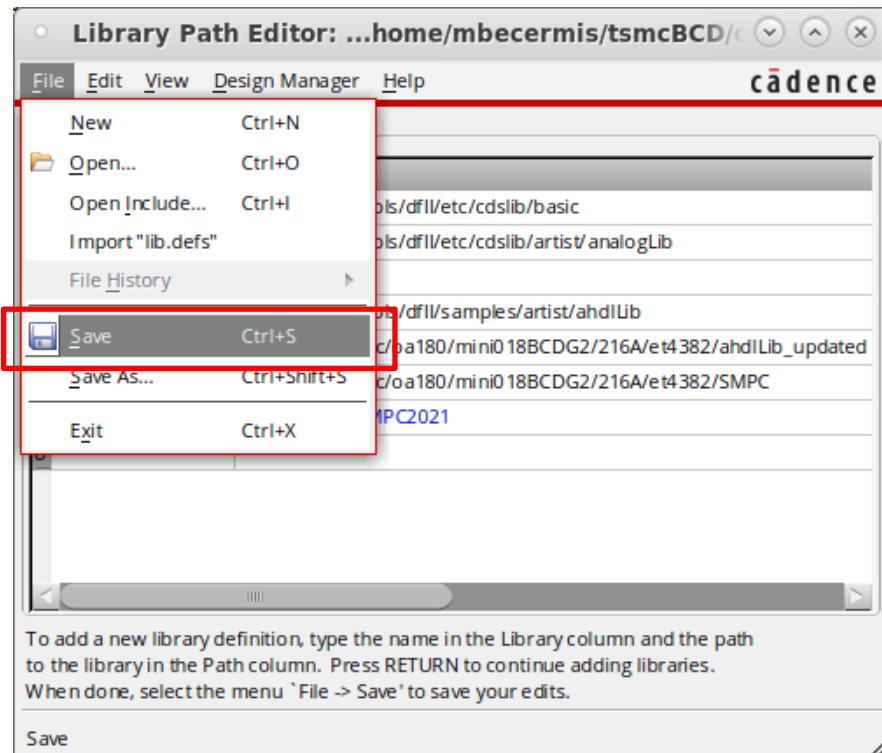


- You can see SMPC2021 library and its path highlighted in blue.

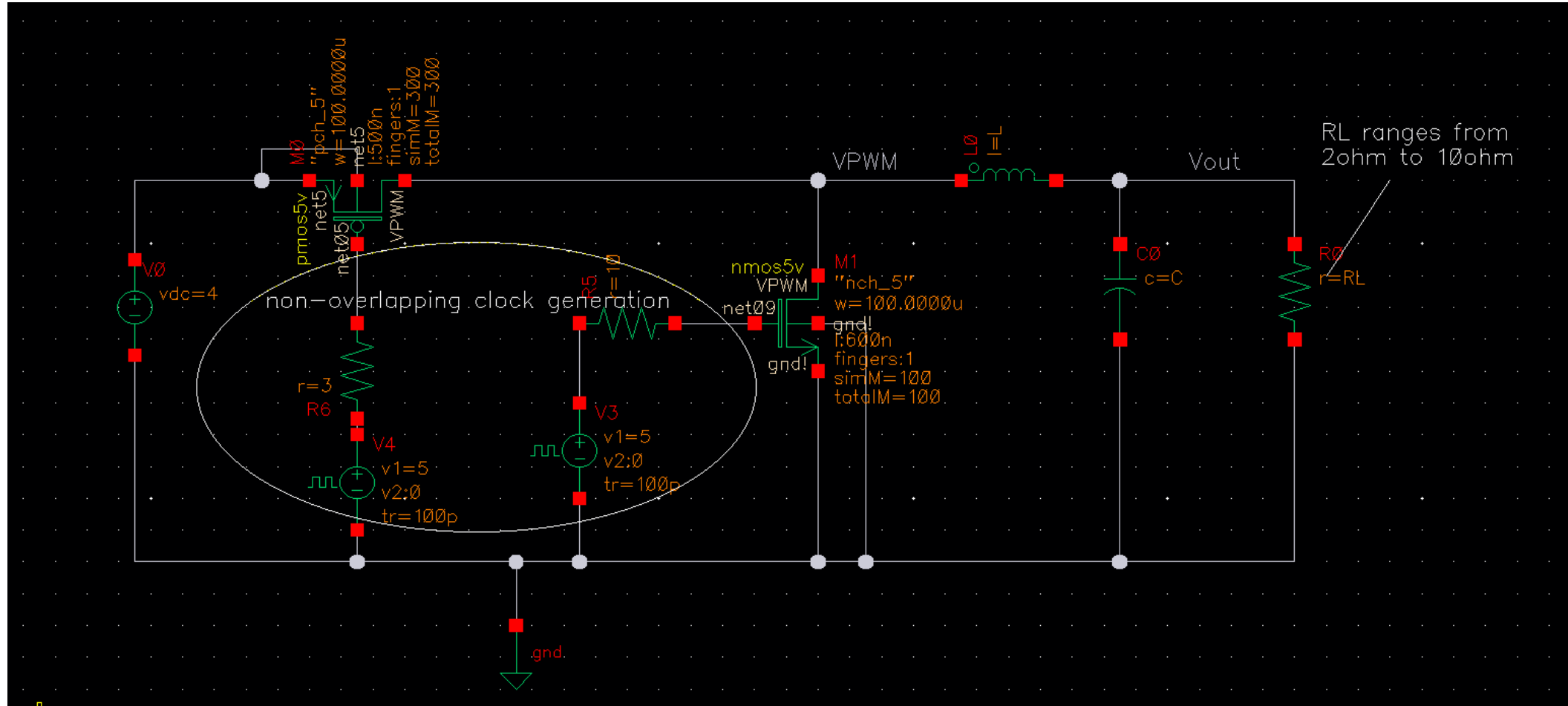
Adding a new library

- **Save** and exit the library path editor.
- You should see SMPC21 in the library manager.
- If you don't, you should refresh the library manager:

Go to **View -> Refresh** in the Library Manager

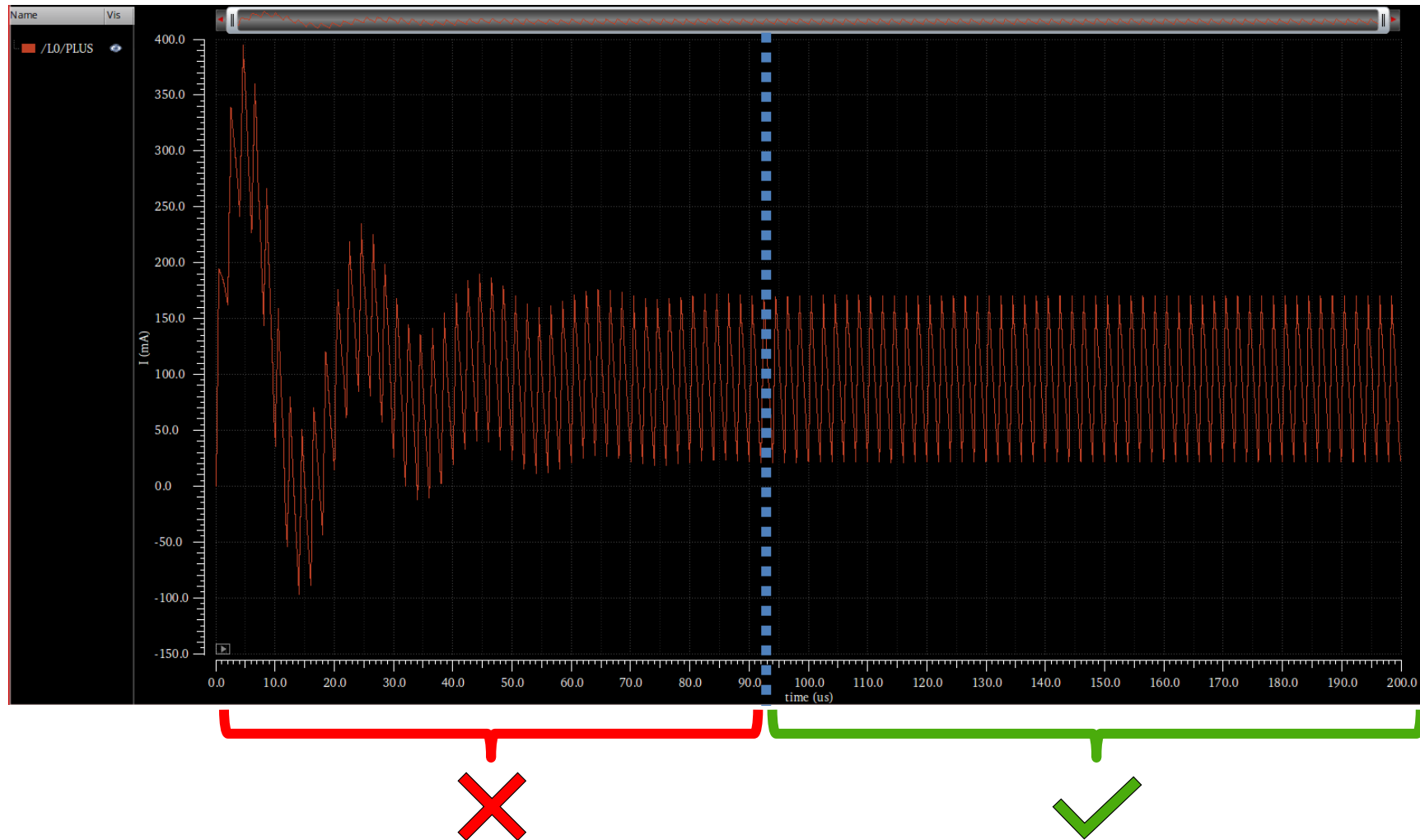


Designing a buck converter



$RL= 10\Omega$, $D=0.25$, $fsw= 500kHz$, $L=10\mu H$, $C=1\mu F$, $tdead=5ns$

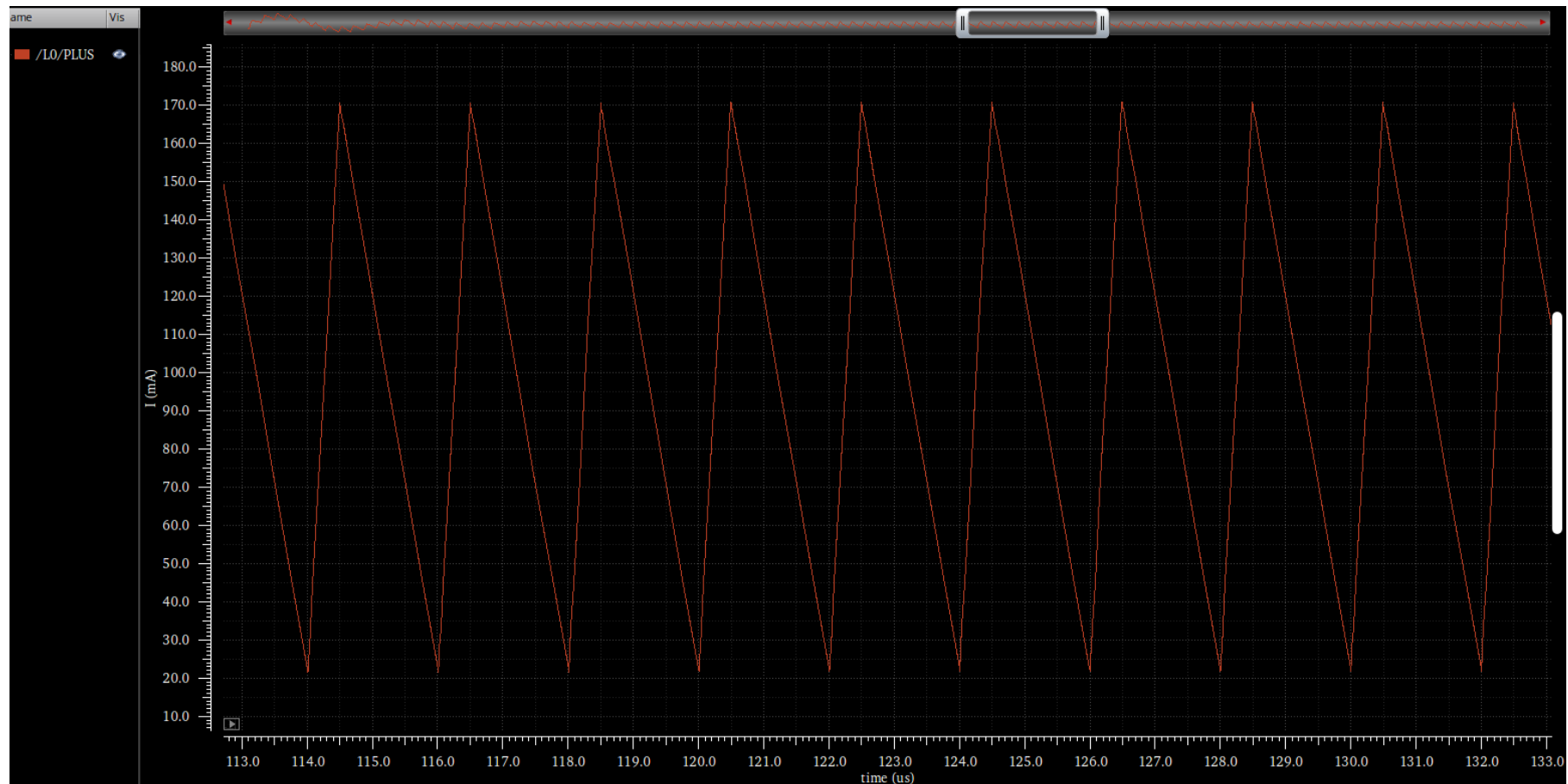
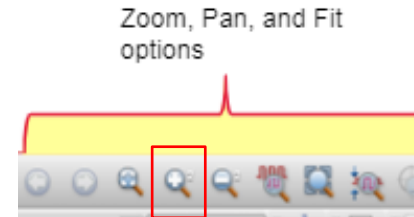
Finding inductor peak-to-peak ripple current with simulations



Do not calculate the inductor current until it fully saturates.

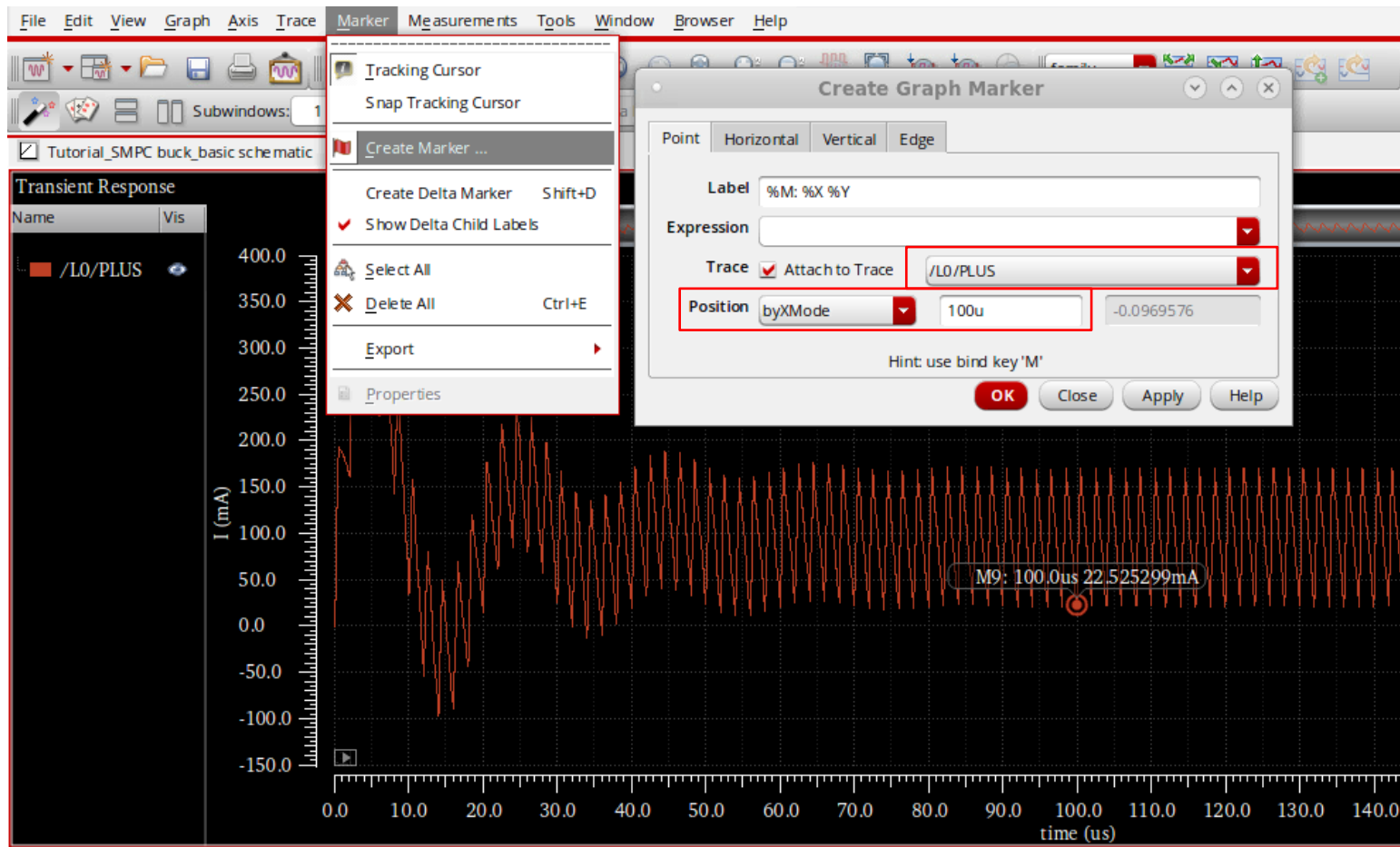
Finding inductor peak-to-peak ripple current with simulations

- To zoom in, either
 - Select the region with right mouse button
 - Use *view toolbar*



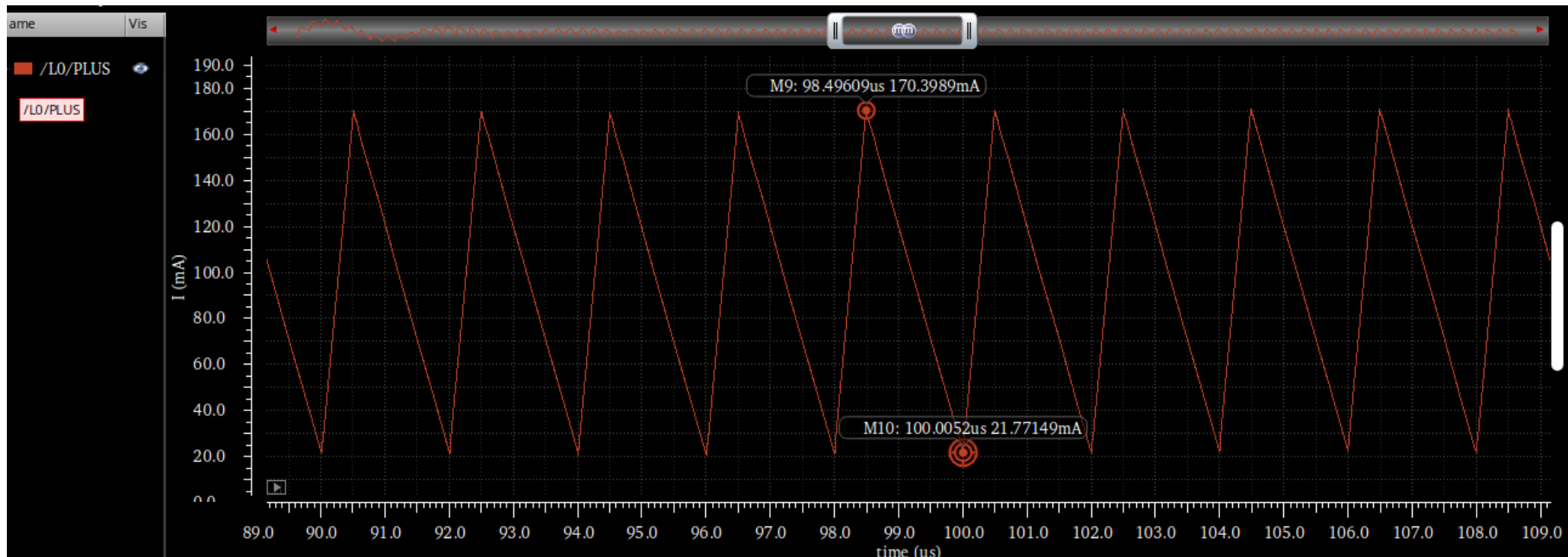
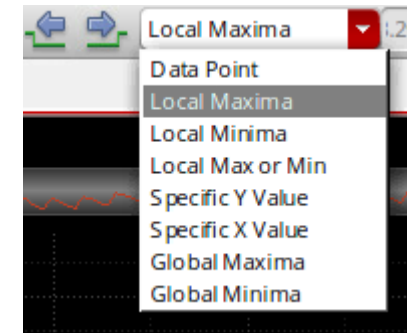
Finding inductor peak-to-peak ripple current with simulations

- Creating a marker either by
 - Pressing **M** button places a marker to where the mouse is.
 - Going to *Marker* -> *Create Marker* and filing “create graph marker”



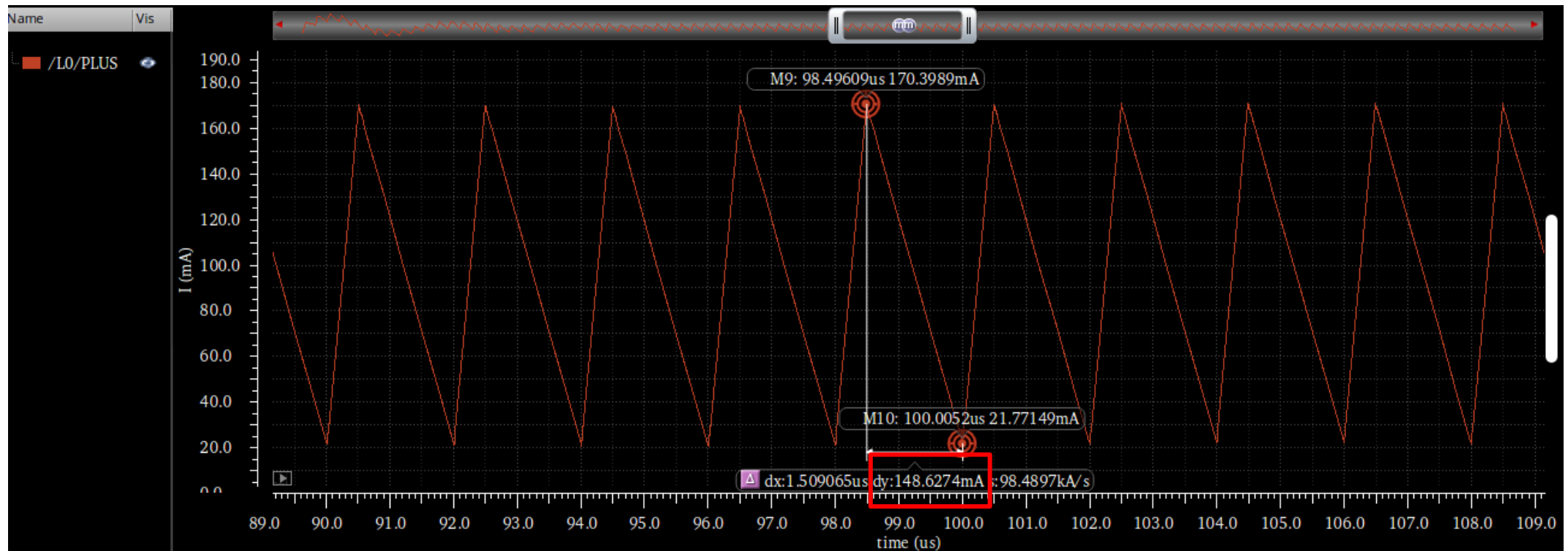
Finding inductor peak-to-peak ripple current with simulations

- To find local maxima and minima use snap tool bar. Snap tool bar moves the selected marker to the local maxima/minima.
- If snap tool bar isn't visible, go to *Window -> Toolbars -> Snap*



Finding the inductor peak-to-peak ripple current with simulations

- To create delta marker
 - Select two markers by holding *Ctrl*
 - Go to *Marker -> Create Delta Marker*
- Hint: After placing and selecting a marker, you can place another marker with **D** button instead of **M**. It will automatically generate a delta marker.



Finding average inductor current with simulations

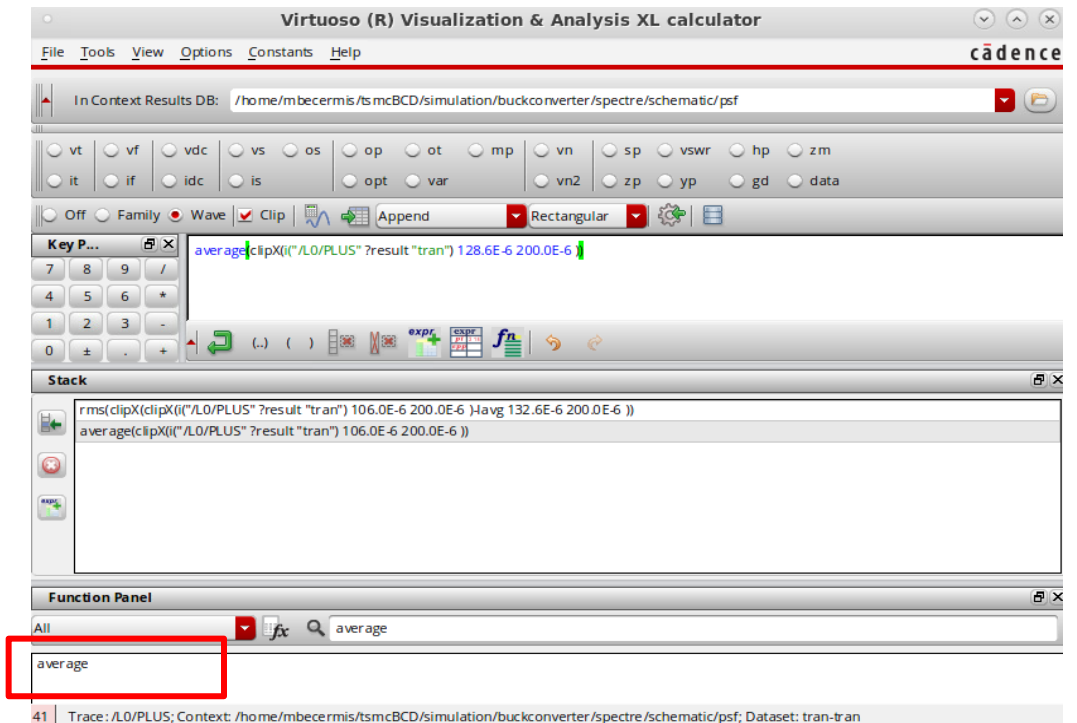
- Open the calculator

- Select Wave



- When you select a signal from the graph, it will be automatically sent to the calculator.
 - If you enable *Clip* option, the trace will be clipped. Otherwise, you need to use *Clip* function to clip the wave.

- Find average function in the function panel.



- Evaluate buffer or send to ADE Outputs



Finding conduction loss with simulations

- Conduction loss can be found by calculating average current and ripple current

- $$P_{cond} = P_{avg} + P_{ripple}$$

- $$P_{avg} = I_{avg}^2 R_{on,pmos} Duty + I_{avg}^2 R_{on,nmos} (1 - Duty)$$

- $$P_{ripple} = I_{ripple,rms}^2 R_{on,pmos} Duty + I_{ripple,rms}^2 R_{on,nmos} (1 - Duty)$$

- Ron can be extracted by using calculator. Find $\frac{V}{I}$ across switches using calculator.
- Plot VT("VPWM")/IT("/M1/D") for NMOS and plot VT("/VPWM")-5)/IT("/M0/D") for PMOS to find ON resistance
- I_{ripple} can be plotted by subtracting I_{avg} from I_{ind} .
- We can take rms of I_{ripple} by using rms function of the calculator.

